

SAE 303

Schemas_reseau_sae303_final(1)(3).drawio

Checklist

0) Pré-requis généraux

- ☐ Plan d'adressage unique et cohérent (aucun doublon)
- ☐ Nommage homogène des VLANs, équipements et interfaces
- ☐ Passerelles des hôtes configurées sur les **VIP HSRP**

1) Administration & sécurité d'accès

Accès SSH

- ☐ SSH activé sur l'ensemble des équipements réseau
(routeurs, switches L3/L2, points d'accès)
- ☐ Accès SSH autorisé **uniquement** depuis le VLAN ADMIN du siège
(10.10.99.0/28)
- ☐ Accès VTY protégé par ACL standard dédiée (MGMT_SSH)

Management des équipements

- ☐ S3-SG (siège) administré via 10.10.99.4/28
- ☐ S1-SC (succursale) administré via 10.20.99.2/28
- ☐ Passerelle de management = VIP ADMIN correspondante

2) Redondance matérielle au siège

Redondance LAN (HSRP sur switches L3)

- ☐ HSRP opérationnel sur les VLANs :
 - SERVICES (10)
 - PERSONNEL (20)
 - VIDEO (30)
 - PRODUCTION (40)
 - WIFI_STAFF (50)
 - WIFI_GUEST (60)
 - ADMIN (99)
- ☐ VIP = .1, S1-SG = .2, S2-SG = .3
- ☐ Bascule fonctionnelle en cas de perte d'un switch L3

Redondance WAN (HSRP sur routeurs)

- ☐ HSRP opérationnel sur le VLAN TRANSIT (100)
 - ☐ VIP = 10.10.100.4
 - ☐ Route par défaut des switches L3 pointant vers la VIP transit
 - ☐ Suivi de lien WAN (track) assurant la bascule automatique
-

3) Services réseau (au siège)

DNS

- ☐ Serveur DNS centralisé dans le VLAN SERVICES (10.10.10.10)
- ☐ Résolution des équipements internes (siège et succursale)
- ☐ Résolution du serveur Web externe

DHCP

- ☐ DHCP centralisé sur le serveur du siège
- ☐ Pools DHCP configurés pour :
 - VLANs du siège (10,20,30,40,50,60)
 - VLANs de la succursale (20,30,40,50,60)
- ☐ Relais DHCP (ip helper-address) configurés sur :
 - Switches L3 du siège
 - Routeur de la succursale
- ☐ VLAN ADMIN en adressage statique (sans DHCP)

Syslog

- ☐ Service Syslog activé sur le serveur du siège
 - ☐ Tous les équipements réseau envoient leurs logs vers le serveur
-

4) Segmentation et filtrage des flux

Proxy HTTP logique (ACL étendues)

- ☐ Filtrage appliqué sur les interfaces WAN des routeurs du siège
- ☐ Accès Internet bloqué pour :
 - VLAN SERVICES
 - VLAN ADMIN
 - VLAN VIDEO
 - VLAN PRODUCTION
- ☐ Accès Internet autorisé :
 - VLAN PERSONNEL : HTTP/HTTPS
 - VLAN WIFI_STAFF : HTTP/HTTPS
 - VLAN WIFI_GUEST : HTTP/HTTPS uniquement
- ☐ Politique par défaut restrictive (deny any)

Segmentation interne

- ☐ Les VLANs utilisateurs sont isolés selon les règles définies
 - ☐ Le VLAN ADMIN reste accessible uniquement pour l'administration
-

5) Réseau Wi-Fi

Siège

- ☐ Point d'accès WIFI_STAFF connecté en VLAN 50
- ☐ Point d'accès WIFI_GUEST connecté en VLAN 60
- ☐ Points d'accès utilisés en mode **pont (AP)** :
 - DHCP désactivé
 - Adresse IP statique dans le VLAN correspondant

Succursale

- ☐ Point d'accès WIFI_STAFF connecté en VLAN 50
 - ☐ Point d'accès WIFI_GUEST connecté en VLAN 60
 - ☐ Même logique de configuration qu'au siège
-

6) Interconnexion Siège ↔ Succursale

- ☐ Tunnel IPsec site-à-site opérationnel
 - ☐ État IPsec validé (QM_IDLE)
 - ☐ Routage cohérent entre les réseaux 10.10.0.0/16 et 10.20.0.0/16
 - ☐ **Full tunnel** :
 - Tout le trafic de la succursale transite par le siège
 - Accès Internet de la succursale soumis au proxy logique du siège
-

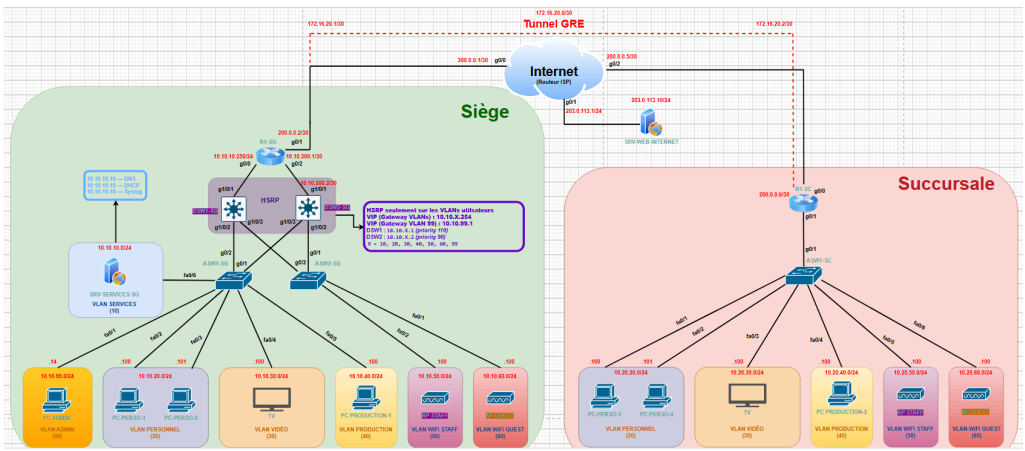
7) Qualité de service (QoS)

- ☐ Classification du trafic VIDEO (VLAN 30)
 - ☐ Priorité élevée appliquée sur les liens WAN
 - ☐ Priorisation effective entre siège et succursale
-

8) Vérifications fonctionnelles

- ☐ Attribution DHCP correcte sur chaque VLAN
- ☐ Résolution DNS fonctionnelle sur tous les sites
- ☐ Accès Internet conforme aux règles de filtrage
- ☐ Bascule HSRP fonctionnelle (LAN et WAN)
- ☐ Flux inter-sites chiffrés et opérationnels
- ☐ Centralisation des logs confirmée

Schémas Réseau et plan d'adressage



Internet caractérisé par un Routeur ISP

WAN / Internet

Lien	Équipement / Interface	IP / Masque	Remarque
ISP ↔ R1-SG	R1-SG g0/0	200.0.0.2/30	GW ISP = 200.0.0.1
ISP ↔ R2-SG	R2-SG g0/0	200.0.0.6/30	GW ISP = 200.0.0.5
ISP ↔ R1-SC	R1-SC g0/2	200.0.0.10/30	GW ISP = 200.0.0.9
ISP ↔ WEB	ISP g0/3	203.0.113.1/24	GW WEB = 203.0.113.1
WEB	WEB NIC	203.0.113.10/24	GW = 203.0.113.1

Siège (LAN 10.10.x.x)

VLANs utilisateurs + passerelles HSRP (sur S1-SG/S2-SG)

VLAN	Nom	Réseau	VIP HSRP (GW des PC)	S1-SG	S2-SG
10	SERVICES	10.10.10.0/24	10.10.10.1	10.10.10.2	10.10.10.3
20	PERSONNEL	10.10.20.0/24	10.10.20.1	10.10.20.2	10.10.20.3
30	VIDEO	10.10.30.0/24	10.10.30.1	10.10.30.2	10.10.30.3
40	PRODUCTION	10.10.40.0/24	10.10.40.1	10.10.40.2	10.10.40.3
50	WIFI_STAFF	10.10.50.0/24	10.10.50.1	10.10.50.2	10.10.50.3
60	WIFI_GUEST	10.10.60.0/24	10.10.60.1	10.10.60.2	10.10.60.3
99	ADMIN (MGMT)	10.10.99.0/28	10.10.99.1	10.10.99.2	10.10.99.3

VLAN 100 Transit (entre L3 et routeurs)

Réseau	Usage	R1-SG	R2-SG	VIP HSRP	S1-SG SVI	S2-SG SVI
10.10.100.0/29	Transit siège	10.10.100.2	10.10.100.3	10.10.100.4	10.10.100.5	10.10.100.6

Management IP du switch L2 (S3-SG)

Équipement	VLAN mgmt	IP	GW
S3-SG	VLAN 99	10.10.99.4/28	10.10.99.1

Serveur (DNS/DHCP/SYSLOG) au siège

Machine	VLAN	IP	GW	DNS
Server-PT (DNS/DHCP/SYSLOG)	VLAN 10	10.10.10.10/24	10.10.10.1	10.10.10.10

Succursale (LAN 10.20.x.x)

VLAN	Nom	Réseau	Gateway (R1-SC subif)
20	PERSONNEL	10.20.20.0/24	10.20.20.1
30	VIDEO	10.20.30.0/24	10.20.30.1
40	PRODUCTION	10.20.40.0/24	10.20.40.1
50	WIFI_STAFF	10.20.50.0/24	10.20.50.1
60	WIFI_GUEST	10.20.60.0/24	10.20.60.1
99	MGMT	10.20.99.0/28	10.20.99.1

Management IP du switch L2 succursale (S1-SC)

Équipement	VLAN mgmt	IP	GW
S1-SC	VLAN 99	10.20.99.2/28	10.20.99.1

WAN / Internet

Configuration Routeur ISP

```
! =====
!  ROUTEUR ISP (WAN)
! =====
hostname ISP
no ip domain-lookup

interface g0/0
description ISP -> R1-SG g0/0
ip address 200.0.0.1 255.255.255.252
no shutdown

interface g0/1
description ISP -> R2-SG g0/0
ip address 200.0.0.5 255.255.255.252
no shutdown
```

```

interface g0/2
description ISP -> R1-SC g0/2
ip address 200.0.0.9 255.255.255.252
no shutdown

interface fa0/0/0
description ISP -> WEB Server (203.0.113.0/24)
ip address 203.0.113.1 255.255.255.0
no shutdown

! =====
! ROUTES (retour vers les LANs)
! =====
! Siège 10.10.0.0/16 : primaire via R1-SG, secours via R2-SG
ip route 10.10.0.0 255.255.0.0 200.0.0.2
ip route 10.10.0.0 255.255.0.0 200.0.0.6 10

! Succursale 10.20.0.0/16 : full-tunnel -> retour via le siège
ip route 10.20.0.0 255.255.0.0 200.0.0.2
ip route 10.20.0.0 255.255.0.0 200.0.0.6 10

```

▼ Configuration WEB Internet

IP du serveur Web (Server-PT)

Desktop → IP Configuration

- IP : 203.0.113.10
- Mask : 255.255.255.0
- Gateway : 203.0.113.1
- DNS : 10.10.10.10

Activer le Web

Services → HTTP → On

(option) Services → HTTPS → On

Siège

Configuration S3-SG (Switch L2)

```

! =====
! S3-SG (L2 ACCESS SWITCH)
! =====

```

```

hostname S3-SG
no ip domain-lookup

vlan 10
  name SERVICES
vlan 20
  name PERSONNEL
vlan 30
  name VIDEO
vlan 40
  name PRODUCTION
vlan 50
  name WIFI_STAFF
vlan 60
  name WIFI_GUEST
vlan 99
  name ADMIN
vlan 100
  name TRANSIT

interface vlan 99
  description MGMT - ADMIN
  ip address 10.10.99.4 255.255.255.240
  no shutdown

ip default-gateway 10.10.99.1

ip domain-name siege.local
username admin privilege 15 secret admin
crypto key generate rsa modulus 1024
ip ssh version 2

ip access-list standard MGMT_SSH
  permit 10.10.99.0 0.0.0.15

line vty 0 4
  login local
  transport input ssh
  access-class MGMT_SSH in

service timestamps log datetime msec
logging host 10.10.10.10
logging trap informational
logging on
logging source-interface vlan 99

spanning-tree mode rapid-pvst
spanning-tree portfast default
spanning-tree bpduguard default

interface g0/1
  description TRUNK -> S1-SG Gi1/0/11
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,40,50,60,99,100

```

```

no shutdown

interface g0/2
description TRUNK -> S2-SG Gi1/0/12
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,99,100
no shutdown

interface f0/1
description SRV-SERVICES
switchport mode access
switchport access vlan 10
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface f0/2
description PC-ADMIN
switchport mode access
switchport access vlan 99
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface f0/3
description PC-PERS0-1
switchport mode access
switchport access vlan 20
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface f0/4
description PC-PERS0-2
switchport mode access
switchport access vlan 20
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface f0/5
description TV-VIDEO
switchport mode access
switchport access vlan 30
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface f0/6
description PC-PRODUCTION
switchport mode access
switchport access vlan 40
spanning-tree portfast
spanning-tree bpduguard enable

```



```

no shutdown

interface f0/7
description AP-STAFF
switchport mode access
switchport access vlan 50
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface f0/8
description AP-GUEST
switchport mode access
switchport access vlan 60
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface range f0/9-24
description UNUSED
shutdown

```

Configuration S1-SG

```

! =====
! S1-SG (L3 CORE - SIEGE) - SANS ACL INTERNES
! =====
hostname S1-SG
no ip domain-lookup
ip routing
standby version 2

vlan 10
name SERVICES
vlan 20
name PERSONNEL
vlan 30
name VIDEO
vlan 40
name PRODUCTION
vlan 50
name WIFI_STAFF
vlan 60
name WIFI_GUEST
vlan 99
name ADMIN
vlan 100
name TRANSIT

ip domain-name siege.local
ip name-server 10.10.10.10

```

```

username admin privilege 15 secret admin
crypto key generate rsa modulus 1024
ip ssh version 2

ip access-list standard MGMT_SSH
permit 10.10.99.0 0.0.0.15

line vty 0 4
login local
transport input ssh
access-class MGMT_SSH in

service timestamps log datetime msec
logging host 10.10.10.10
logging trap informational
logging on
logging source-interface vlan 99

interface vlan 100
description Transit vers routeurs
ip address 10.10.100.5 255.255.255.248
no shutdown

interface vlan 10
description SERVICES
ip address 10.10.10.2 255.255.255.0
standby 10 ip 10.10.10.1
standby 10 priority 110
standby 10 preempt
no shutdown

interface vlan 20
description PERSONNEL
ip address 10.10.20.2 255.255.255.0
standby 20 ip 10.10.20.1
standby 20 priority 110
standby 20 preempt
ip helper-address 10.10.10.10
no shutdown

interface vlan 30
description VIDEO
ip address 10.10.30.2 255.255.255.0
standby 30 ip 10.10.30.1
standby 30 priority 110
standby 30 preempt
ip helper-address 10.10.10.10
no shutdown

interface vlan 40
description PRODUCTION
ip address 10.10.40.2 255.255.255.0
standby 40 ip 10.10.40.1

```

```

standby 40 priority 110
standby 40 preempt
ip helper-address 10.10.10.10
no shutdown

interface vlan 50
description WIFI_STAFF
ip address 10.10.50.2 255.255.255.0
standby 50 ip 10.10.50.1
standby 50 priority 110
standby 50 preempt
ip helper-address 10.10.10.10
no shutdown

interface vlan 60
description WIFI_GUEST
ip address 10.10.60.2 255.255.255.0
standby 60 ip 10.10.60.1
standby 60 priority 110
standby 60 preempt
ip helper-address 10.10.10.10
no shutdown

interface vlan 99
description ADMIN
ip address 10.10.99.2 255.255.255.240
standby 99 ip 10.10.99.1
standby 99 priority 110
standby 99 preempt
no shutdown

! Route par défaut vers la VIP HSRP des routeurs
ip route 0.0.0.0 0.0.0.0 10.10.100.4

interface gil/0/1
description Vers R1-SG g0/1 (VLAN100)
switchport mode access
switchport access vlan 100
no shutdown

interface gil/0/11
description Vers S3-SG (TRUNK)
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,99,100
no shutdown

```

Configuration S2-SG

```

! =====
! S2-SG (L3 CORE - SIEGE) - SANS ACL INTERNES
! =====

```

```

hostname S2-SG
no ip domain-lookup
ip routing
standby version 2

vlan 10
  name SERVICES
vlan 20
  name PERSONNEL
vlan 30
  name VIDEO
vlan 40
  name PRODUCTION
vlan 50
  name WIFI_STAFF
vlan 60
  name WIFI_GUEST
vlan 99
  name ADMIN
vlan 100
  name TRANSIT

ip domain-name siege.local
ip name-server 10.10.10.10

username admin privilege 15 secret admin
crypto key generate rsa modulus 1024
ip ssh version 2

ip access-list standard MGMT_SSH
  permit 10.10.99.0 0.0.0.15

line vty 0 4
  login local
  transport input ssh
  access-class MGMT_SSH in

service timestamps log datetime msec
logging host 10.10.10.10
logging trap informational
logging on
logging source-interface vlan 99

interface vlan 100
  description Transit vers routeurs
  ip address 10.10.100.6 255.255.255.248
  no shutdown

interface vlan 10
  description SERVICES
  ip address 10.10.10.3 255.255.255.0
  standby 10 ip 10.10.10.1
  standby 10 priority 90
  standby 10 preempt

```

```

no shutdown

interface vlan 20
description PERSONNEL
ip address 10.10.20.3 255.255.255.0
standby 20 ip 10.10.20.1
standby 20 priority 90
standby 20 preempt
ip helper-address 10.10.10.10
no shutdown

interface vlan 30
description VIDEO
ip address 10.10.30.3 255.255.255.0
standby 30 ip 10.10.30.1
standby 30 priority 90
standby 30 preempt
ip helper-address 10.10.10.10
no shutdown

interface vlan 40
description PRODUCTION
ip address 10.10.40.3 255.255.255.0
standby 40 ip 10.10.40.1
standby 40 priority 90
standby 40 preempt
ip helper-address 10.10.10.10
no shutdown

interface vlan 50
description WIFI_STAFF
ip address 10.10.50.3 255.255.255.0
standby 50 ip 10.10.50.1
standby 50 priority 90
standby 50 preempt
ip helper-address 10.10.10.10
no shutdown

interface vlan 60
description WIFI_GUEST
ip address 10.10.60.3 255.255.255.0
standby 60 ip 10.10.60.1
standby 60 priority 90
standby 60 preempt
ip helper-address 10.10.10.10
no shutdown

interface vlan 99
description ADMIN
ip address 10.10.99.3 255.255.255.240
standby 99 ip 10.10.99.1
standby 99 priority 90
standby 99 preempt
no shutdown

```

```

! Route par défaut vers la VIP HSRP des routeurs
ip route 0.0.0.0 0.0.0.0 10.10.100.4

interface gil/0/1
description Vers R2-SG g0/1 (VLAN100)
switchport mode access
switchport access vlan 100
no shutdown

interface gil/0/12
description Vers S3-SG (TRUNK)
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,99,100
no shutdown

```

Configuration R1-SG

```

! =====
! R1-SG (SIEGE - PRIMARY)
! =====
hostname R1-SG
no ip domain-lookup

ip domain-name siege.local
ip name-server 10.10.10.10

enable secret admin
username admin privilege 15 secret admin
crypto key generate rsa modulus 1024
ip ssh version 2

ip access-list standard MGMT_SSH
permit 10.10.99.0 0.0.0.15

line vty 0 4
login local
transport input ssh
access-class MGMT_SSH in

service timestamps log datetime msec
logging host 10.10.10.10
logging trap informational
logging on
logging source-interface g0/1

crypto isakmp identity address
crypto isakmp nat keepalive 20

! =====
! IPsec (Full-tunnel Succursale)

```

```

! =====
ip access-list extended VPN-TRAFFIC
  remark --- Inter-site LAN <-> LAN ---
  permit ip 10.10.0.0 0.0.255.255 10.20.0.0 0.0.255.255
  remark --- Full tunnel : Succursale Internet via Siege ---
  permit ip any 10.20.0.0 0.0.255.255

crypto isakmp policy 10
  encryption aes
  hash sha
  authentication pre-share
  group 2
  lifetime 86400

crypto isakmp key VPNKEY address 200.0.0.10
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac

crypto map VPN-MAP 10 ipsec-isakmp
  set peer 200.0.0.10
  set transform-set VPN-SET
  match address VPN-TRAFFIC

! =====
! QoS VIDEO (VLAN 30) - PRIORITE FORTE
! (utile aussi pour inter-site grâce à "qos pre-classify")
! =====
ip access-list extended QOS_VIDEO_MATCH
  remark --- VLAN VIDEO Siege + Succursale ---
  permit ip 10.10.30.0 0.0.0.255 any
  permit ip 10.20.30.0 0.0.0.255 any

class-map match-any CM_VIDEO
  match access-group name QOS_VIDEO_MATCH

policy-map PM_WAN_OUT
  class CM_VIDEO
    set dscp af41
    priority percent 40
  class class-default
    fair-queue

! =====
! PROXY WEB (ACL WAN OUT)
! =====
ip access-list extended PROXY_LOGIQUE
  remark --- Autoriser INTER-SITE (VPN) ---
  permit ip 10.10.0.0 0.0.255.255 10.20.0.0 0.0.255.255
  permit ip 10.20.0.0 0.0.255.255 10.10.0.0 0.0.255.255

  remark --- IPsec/IKE/NAT-T vers Succursale ---
  permit udp host 200.0.0.2 host 200.0.0.10 eq 500
  permit udp host 200.0.0.2 host 200.0.0.10 eq 4500
  permit esp host 200.0.0.2 host 200.0.0.10

```

```

remark --- DNS externe autorisé UNIQUEMENT depuis le serveur DNS ---
permit udp host 10.10.10.10 any eq 53
permit tcp host 10.10.10.10 any eq 53

remark --- BLOQUER Internet pour SERVICES (VLAN10) ---
deny ip 10.10.10.0 0.0.0.255 any log

remark --- BLOQUER Internet pour ADMIN (VLAN99 /28) ---
deny ip 10.10.99.0 0.0.0.15 any log

remark --- GUEST : Web uniquement (80/443) ---
permit tcp 10.10.60.0 0.0.0.255 any eq 80
permit tcp 10.10.60.0 0.0.0.255 any eq 443
permit tcp 10.20.60.0 0.0.0.255 any eq 80
permit tcp 10.20.60.0 0.0.0.255 any eq 443

remark --- PERSONNEL + STAFF : Web (80/443) ---
permit tcp 10.10.20.0 0.0.0.255 any eq 80
permit tcp 10.10.20.0 0.0.0.255 any eq 443
permit tcp 10.10.50.0 0.0.0.255 any eq 80
permit tcp 10.10.50.0 0.0.0.255 any eq 443
permit tcp 10.20.20.0 0.0.0.255 any eq 80
permit tcp 10.20.20.0 0.0.0.255 any eq 443
permit tcp 10.20.50.0 0.0.0.255 any eq 80
permit tcp 10.20.50.0 0.0.0.255 any eq 443

remark --- BLOQUER Internet pour VIDEO + PRODUCTION ---
deny ip 10.10.30.0 0.0.0.255 any log
deny ip 10.10.40.0 0.0.0.255 any log
deny ip 10.20.30.0 0.0.0.255 any log
deny ip 10.20.40.0 0.0.0.255 any log

deny ip any any log

! =====
! NAT/PAT (full-tunnel)
! =====
ip access-list extended NAT_EXEMPT
permit ip 10.10.0.0 0.0.255.255 10.20.0.0 0.0.255.255
permit ip 10.20.0.0 0.0.255.255 10.10.0.0 0.0.255.255

ip access-list extended NAT_INTERNET
permit ip 10.10.0.0 0.0.255.255 any
permit ip 10.20.0.0 0.0.255.255 any

route-map RM_NAT deny 10
match ip address NAT_EXEMPT
route-map RM_NAT permit 20
match ip address NAT_INTERNET

! =====
! INTERFACES
! =====
interface g0/0

```



```

description WAN -> ISP
ip address 200.0.0.2 255.255.255.252
qos pre-classify
service-policy output PM_WAN_OUT
ip access-group PROXY_LOGIQUE out
crypto map VPN-MAP
crypto isakmp enable
ip nat outside
no shutdown

interface g0/1
description Transit VLAN100 -> L3 (HSRP)
ip address 10.10.100.2 255.255.255.248
standby version 2
standby 100 ip 10.10.100.4
standby 100 priority 110
standby 100 preempt
standby 100 track g0/0 30
ip nat inside
no shutdown

! NAT overload sur WAN
ip nat inside source route-map RM_NAT interface g0/0 overload

! =====
! ROUTING
! =====
ip route 0.0.0.0 0.0.0.0 200.0.0.1
ip route 200.0.0.10 255.255.255.255 200.0.0.1

```

Configuration R2-SG

```

! =====
! R2-SG (SIEGE - BACKUP)
! =====
hostname R2-SG
no ip domain-lookup

ip domain-name siege.local
ip name-server 10.10.10.10

enable secret admin
username admin privilege 15 secret admin
crypto key generate rsa modulus 1024
ip ssh version 2

ip access-list standard MGMT_SSH
permit 10.10.99.0 0.0.0.15

line vty 0 4
login local

```

```

transport input ssh
access-class MGMT_SSH in

service timestamps log datetime msec
logging host 10.10.10.10
logging trap informational
logging on
logging source-interface g0/1

crypto isakmp identity address
crypto isakmp nat keepalive 20

! =====
! IPsec : backup headend (Full-tunnel Succursale)
! =====
ip access-list extended VPN-TRAFFIC
 remark --- Inter-site LAN <-> LAN ---
 permit ip 10.10.0.0 0.0.255.255 10.20.0.0 0.0.255.255
 remark --- Full tunnel : Succursale Internet via Siege ---
 permit ip any 10.20.0.0 0.0.255.255

crypto isakmp policy 10
 encryption aes
 hash sha
 authentication pre-share
 group 2
 lifetime 86400

crypto isakmp key VPNKEY address 200.0.0.10
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac

crypto map VPN-MAP 10 ipsec-isakmp
 set peer 200.0.0.10
 set transform-set VPN-SET
 match address VPN-TRAFFIC

! =====
! QoS VIDEO (VLAN 30) - PRIORITE FORTE
! =====
ip access-list extended QOS_VIDEO_MATCH
 remark --- VLAN VIDEO Siege + Succursale ---
 permit ip 10.10.30.0 0.0.0.255 any
 permit ip 10.20.30.0 0.0.0.255 any

class-map match-any CM_VIDEO
 match access-group name QOS_VIDEO_MATCH

policy-map PM_WAN_OUT
 class CM_VIDEO
  set dscp af41
  priority percent 40
 class class-default
  fair-queue

```

```

! =====
! PROXY WEB (ACL WAN OUT)
! =====
ip access-list extended PROXY_LOGIQUE
remark --- Autoriser INTER-SITE (VPN) ---
permit ip 10.10.0.0 0.0.255.255 10.20.0.0 0.0.255.255
permit ip 10.20.0.0 0.0.255.255 10.10.0.0 0.0.255.255

remark --- IPsec/IKE/NAT-T vers Succursale ---
permit udp host 200.0.0.6 host 200.0.0.10 eq 500
permit udp host 200.0.0.6 host 200.0.0.10 eq 4500
permit esp host 200.0.0.6 host 200.0.0.10

remark --- DNS externe autorisé UNIQUEMENT depuis le serveur DNS ---
permit udp host 10.10.10.10 any eq 53
permit tcp host 10.10.10.10 any eq 53

remark --- BLOQUER Internet pour SERVICES (VLAN10) ---
deny ip 10.10.10.0 0.0.0.255 any log

remark --- BLOQUER Internet pour ADMIN (VLAN99 /28) ---
deny ip 10.10.99.0 0.0.0.15 any log

remark --- GUEST : Web uniquement (80/443) ---
permit tcp 10.10.60.0 0.0.0.255 any eq 80
permit tcp 10.10.60.0 0.0.0.255 any eq 443
permit tcp 10.20.60.0 0.0.0.255 any eq 80
permit tcp 10.20.60.0 0.0.0.255 any eq 443

remark --- PERSONNEL + STAFF : Web (80/443) ---
permit tcp 10.10.20.0 0.0.0.255 any eq 80
permit tcp 10.10.20.0 0.0.0.255 any eq 443
permit tcp 10.10.50.0 0.0.0.255 any eq 80
permit tcp 10.10.50.0 0.0.0.255 any eq 443
permit tcp 10.20.20.0 0.0.0.255 any eq 80
permit tcp 10.20.20.0 0.0.0.255 any eq 443
permit tcp 10.20.50.0 0.0.0.255 any eq 80
permit tcp 10.20.50.0 0.0.0.255 any eq 443

remark --- BLOQUER Internet pour VIDEO + PRODUCTION ---
deny ip 10.10.30.0 0.0.0.255 any log
deny ip 10.10.40.0 0.0.0.255 any log
deny ip 10.20.30.0 0.0.0.255 any log
deny ip 10.20.40.0 0.0.0.255 any log

deny ip any any log

! =====
! NAT/PAT (backup)
! =====
ip access-list extended NAT_EXEMPT
permit ip 10.10.0.0 0.0.255.255 10.20.0.0 0.0.255.255
permit ip 10.20.0.0 0.0.255.255 10.10.0.0 0.0.255.255

```

```

ip access-list extended NAT_INTERNET
  permit ip 10.10.0.0 0.0.255.255 any
  permit ip 10.20.0.0 0.0.255.255 any

route-map RM_NAT deny 10
  match ip address NAT_EXEMPT
route-map RM_NAT permit 20
  match ip address NAT_INTERNET

! =====
! INTERFACES
! =====
interface g0/0
  description WAN -> ISP
  ip address 200.0.0.6 255.255.255.252
  qos pre-classify
  service-policy output PM_WAN_OUT
  ip access-group PROXY_LOGIQUE out
  crypto map VPN-MAP
  crypto isakmp enable
  ip nat outside
  no shutdown

interface g0/1
  description Transit VLAN100 -> L3 (HSRP)
  ip address 10.10.100.3 255.255.255.248
  standby version 2
  standby 100 ip 10.10.100.4
  standby 100 priority 90
  standby 100 preempt
  standby 100 track g0/0 30
  ip nat inside
  no shutdown

! NAT overload sur WAN
ip nat inside source route-map RM_NAT interface g0/0 overload

! =====
! ROUTING
! =====
ip route 0.0.0.0 0.0.0.0 200.0.0.5
ip route 200.0.0.10 255.255.255.255 200.0.0.5

```

▼ Configuration du DNS

DNS (au siège, VLAN 10 SERVICES) – à faire sur Packet Tracer

1) Serveur DNS (Server-PT)

- Branché sur **S3-SG** sur un port **access VLAN 10**
- IP du serveur :
 - IP **10.10.10.10**

- **Mask** 255.255.255.0
- **GW** 10.10.10.1 (VIP HSRP VLAN 10)
- **DNS** 10.10.10.10

2) Activer DNS

Sur le serveur : **Services** → **DNS** → **On**

3) Ajouter les entrées DNS (A records)

Siège

- **DNS-SG** → 10.10.10.10
- **S1-SG** → 10.10.99.2
- **S2-SG** → 10.10.99.3
- **S3-SG** → 10.10.99.4
- **R1-SG** → 10.10.100.2
- **R2-SG** → 10.10.100.3

Succursale (résolution OK, ping/ssh seulement quand IPsec sera up)

- **R1-SC** → 10.20.99.1
- **S1-SC** → 10.20.99.2

WAN

- **WEB** → 203.0.113.10
- **ISP-WEB** → 203.0.113.1

4) Dire à tout le monde d'utiliser ce DNS

- Sur **PC ADMIN** et les autres hôtes : DNS = 10.10.10.10

▼ Configuration du DHCP (sur le même Server-PT que DNS)

1) Server-PT (DNS/DHCP) – IP fixe

Desktop → IP Configuration

- IP : 10.10.10.10
- Mask : 255.255.255.0
- Gateway : 10.10.10.1
- DNS : 10.10.10.10

2) DHCP sur Server-PT (Packet Tracer)

Services → DHCP → On

Crée ces pools (un par VLAN).

✓ SIÈGE

Pool : SG_SERVICES (VLAN10)

- Default Gateway : 10.10.10.1
- DNS Server : 10.10.10.10
- Start IP : 10.10.10.100
- Subnet Mask : 255.255.255.0
- Maximum Users : 100

Pool : SG_PERSONNEL (VLAN20)

- GW : 10.10.20.1
- DNS : 10.10.10.10
- Start : 10.10.20.100
- Mask : 255.255.255.0
- Max : 100

Pool : SG_VIDEO (VLAN30)

- GW : 10.10.30.1
- DNS : 10.10.10.10
- Start : 10.10.30.100
- Mask : 255.255.255.0
- Max : 100

Pool : SG_PRODUCTION (VLAN40)

- GW : 10.10.40.1
- DNS : 10.10.10.10
- Start : 10.10.40.100
- Mask : 255.255.255.0
- Max : 100

Pool : SG_WIFI_STAFF (VLAN50)

- GW : 10.10.50.1
- DNS : 10.10.10.10
- Start : 10.10.50.100
- Mask : 255.255.255.0
- Max : 100

Pool : SG_WIFI_GUEST (VLAN60)

- GW : 10.10.60.1
- DNS : 10.10.10.10
- Start : 10.10.60.100
- Mask : 255.255.255.0
- Max : 100

✓ SUCCURSALE (*plans en 10.20.x.x*)

Pool : SC_PERSONNEL (VLAN20)

- GW : 10.20.20.1
- DNS : 10.10.10.10
- Start : 10.20.20.100
- Mask : 255.255.255.0
- Max : 100

Pool : SC_VIDEO (VLAN30)

- GW : 10.20.30.1
- DNS : 10.10.10.10
- Start : 10.20.30.100
- Mask : 255.255.255.0
- Max : 100

Pool : SC_PRODUCTION (VLAN40)

- GW : 10.20.40.1
- DNS : 10.10.10.10
- Start : 10.20.40.100
- Mask : 255.255.255.0
- Max : 100

Pool : SC_WIFI_STAFF (VLAN50)

- GW : 10.20.50.1
- DNS : 10.10.10.10
- Start : 10.20.50.100
- Mask : 255.255.255.0
- Max : 100

Pool : SC_WIFI_GUEST (VLAN60)

- GW : 10.20.60.1
- DNS : 10.10.10.10
- Start : 10.20.60.100
- Mask : 255.255.255.0
- Max : 100

▼ Configuration de SYSLOG (sur le même Server-PT que DNS et DHCP)

Activer SYSLOG

Services → SYSLOG → On

→ C'est tout côté serveur. Les logs vont apparaître dans cette page.

▼ Configuration Wi-Fi STAFF

AP STAFF – Siège

Équipement

- **Wireless Router (WRT300N)**
(utilisé **comme point d'accès**, pas comme routeur)

Branchement

- **Port LAN** du WRT300N → **S3-SG**
- Le port du switch est **déjà en VLAN 50**

Configuration du WRT300N

Desktop → IP Configuration

- IP : **10.10.50.10**
- Mask : **255.255.255.0**
- Gateway : **10.10.50.1**
- DHCP : **OFF**

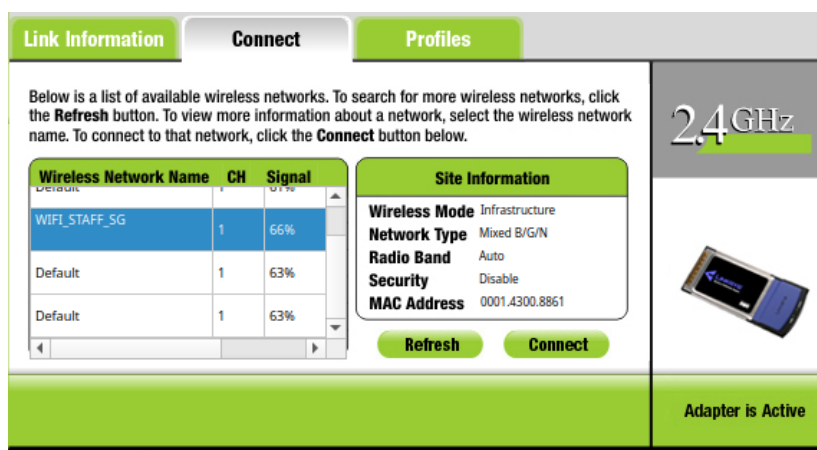
Wireless

- SSID : **WIFI_STAFF_SG**

Wireless Security

- **Security Mode** : **WPA2-PSK**
- **Encryption** : **AES**
- **Passphrase** :

StaffWifi@2026



▼ Configuration Wi-Fi GUEST

AP GUEST – Siège

Équipement

- Wireless Router (WRT300N)

Branchement

- Port LAN du WRT300N → S3-SG
- Le port du switch est déjà en VLAN 60

Configuration du WRT300N

Desktop → IP Configuration

- IP : 10.10.60.10
- Mask : 255.255.255.0
- Gateway : 10.10.60.1
- DHCP : OFF

Wireless

- SSID : WIFI_GUEST_SG

Wireless Security

- Security Mode : WPA2-PSK
- Encryption : AES
- Passphrase :

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▼ Flux Vidéo

Aide : [TV configuration in Cisco packet tracer by || Prashant Ardeshana ||](#)

Succursale

Configuration de R1-SC

```
! =====  
! R1-SC (Routeur Succursale)  
! =====  
hostname R1-SC  
no ip domain-lookup
```

```

ip domain-name succursale.local
ip name-server 10.10.10.10

enable secret admin
username admin privilege 15 secret admin
crypto key generate rsa modulus 1024
ip ssh version 2

! SSH uniquement depuis ADMIN du siège (10.10.99.0/28)
ip access-list standard MGMT_SSH
  permit 10.10.99.0 0.0.0.15

line vty 0 4
  login local
  transport input ssh
  access-class MGMT_SSH in

service timestamps log datetime msec
logging host 10.10.10.10
logging trap informational
logging on
logging source-interface g0/1.99

crypto isakmp identity address
crypto isakmp nat keepalive 20

! =====
! QoS VIDEO (VLAN 30) - PRIORITE FORTE (Succursale -> WAN)
! =====
ip access-list extended QOS_VIDEO_MATCH
  remark --- VLAN VIDEO Succursale (10.20.30.0/24) ---
  permit ip 10.20.30.0 0.0.0.255 any

class-map match-any CM_VIDEO
  match access-group name QOS_VIDEO_MATCH

policy-map PM_WAN_OUT
  class CM_VIDEO
    set dscp af41
    priority percent 40
  class class-default
    fair-queue

! =====
! WAN vers ISP
! =====
interface g0/2
  description WAN -> ISP
  ip address 200.0.0.10 255.255.255.252
  qos pre-classify
  service-policy output PM_WAN_OUT
  crypto map VPN-MAP
  crypto isakmp enable

```

```

no shutdown

! =====
! TRUNK vers switch succursale
! =====
interface g0/1
description TRUNK -> S1-SC g0/1
no shutdown

interface g0/1.20
encapsulation dot1Q 20
description VLAN20 PERSONNEL
ip address 10.20.20.1 255.255.255.0
ip helper-address 10.10.10.10

interface g0/1.30
encapsulation dot1Q 30
description VLAN30 VIDEO
ip address 10.20.30.1 255.255.255.0
ip helper-address 10.10.10.10

interface g0/1.40
encapsulation dot1Q 40
description VLAN40 PRODUCTION
ip address 10.20.40.1 255.255.255.0
ip helper-address 10.10.10.10

interface g0/1.50
encapsulation dot1Q 50
description VLAN50 WIFI_STAFF
ip address 10.20.50.1 255.255.255.0
ip helper-address 10.10.10.10

interface g0/1.60
encapsulation dot1Q 60
description VLAN60 WIFI_GUEST
ip address 10.20.60.1 255.255.255.0
ip helper-address 10.10.10.10

interface g0/1.99
encapsulation dot1Q 99
description VLAN99 MGMT (IPs statiques)
ip address 10.20.99.1 255.255.255.240

! =====
! ROUTING (FULL-TUNNEL)
! =====
ip route 0.0.0.0 0.0.0.0 200.0.0.9
ip route 200.0.0.2 255.255.255.255 200.0.0.9
ip route 200.0.0.6 255.255.255.255 200.0.0.9

! =====
! IPsec – trafic intéressant
! =====

```

```

ip access-list extended VPN-TRAFFIC
  remark --- Inter-site LAN <-> LAN ---
  permit ip 10.20.0.0 0.0.255.255 10.10.0.0 0.0.255.255
  remark --- Full tunnel : Internet via siege ---
  permit ip 10.20.0.0 0.0.255.255 any

crypto isakmp policy 10
  encryption aes
  hash sha
  authentication pre-share
  group 2
  lifetime 86400

crypto isakmp key VPNKEY address 200.0.0.2
crypto isakmp key VPNKEY address 200.0.0.6

crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac

crypto map VPN-MAP 10 ipsec-isakmp
  set peer 200.0.0.2
  set transform-set VPN-SET
  match address VPN-TRAFFIC

crypto map VPN-MAP 20 ipsec-isakmp
  set peer 200.0.0.6
  set transform-set VPN-SET
  match address VPN-TRAFFIC

```

Configuration S1-SC

```

! =====
! S1-SC (L2 ACCESS SWITCH - Succursale)
! =====
hostname S1-SC
no ip domain-lookup

vlan 20
  name PERSONNEL
vlan 30
  name VIDEO
vlan 40
  name PRODUCTION
vlan 50
  name WIFI_STAFF
vlan 60
  name WIFI_GUEST
vlan 99
  name MGMT

interface vlan 99
  description MGMT - Switch

```

```

ip address 10.20.99.2 255.255.255.240
no shutdown

ip default-gateway 10.20.99.1

ip domain-name succursale.local
ip name-server 10.10.10.10

username admin privilege 15 secret admin
crypto key generate rsa modulus 1024
ip ssh version 2

ip access-list standard MGMT_SSH
permit 10.10.99.0 0.0.0.15

line vty 0 4
login local
transport input ssh
access-class MGMT_SSH in

service timestamps log datetime msec
logging host 10.10.10.10
logging trap informational
logging on
logging source-interface vlan 99

spanning-tree mode rapid-pvst
spanning-tree portfast default
spanning-tree bpduguard default

interface g0/1
description TRUNK -> R1-SC g0/1
switchport mode trunk
switchport trunk allowed vlan 20,30,40,50,60,99
no shutdown

interface fa0/1
description PC-PERS0-3
switchport mode access
switchport access vlan 20
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface fa0/2
description PC-PERS0-4
switchport mode access
switchport access vlan 20
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface fa0/3
description TV

```

```

switchport mode access
switchport access vlan 30
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface fa0/4
description PC-PRODUCTION-2
switchport mode access
switchport access vlan 40
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface fa0/5
description AP-STAFF
switchport mode access
switchport access vlan 50
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface fa0/6
description AP-GUEST
switchport mode access
switchport access vlan 60
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown

interface range fa0/7-24
description UNUSED
shutdown

```

▼ Configuration Wi-Fi STAFF

AP STAFF – Succursale

Équipement

- Wireless Router (WRT300N)

Branchement

- Port LAN du WRT300N → S1-SC
- Le port du switch est **déjà en VLAN 50**

Configuration du WRT300N

Desktop → IP Configuration

- IP : 10.20.50.10
- Mask : 255.255.255.0

- Gateway : 10.20.50.1
- DHCP : OFF

Wireless

- SSID : WIFI_STAFF_SC

Wireless Security

- WPA2-PSK

▼ Configuration Wi-Fi GUEST

AP GUEST – Succursale

Équipement

- Wireless Router (WRT300N)

Branchement

- Port LAN du WRT300N → S1-SC
- Le port du switch est déjà en VLAN 60

Configuration du WRT300N

Desktop → IP Configuration

- IP : 10.20.60.10
- Mask : 255.255.255.0
- Gateway : 10.20.60.1
- DHCP : OFF

Wireless

- SSID : WIFI_GUEST_SC

Wireless Security

- Open ou WPA2

▼ Flux Vidéo

Aide : [TV configuration in Cisco packet tracer by || Prashant Ardeshana ||](#)

▼ Configuration avec ACL internes pour séparer les VLANs (a faire que si tout fonctionne)

Configuration de S1-SG

```
! =====
! S1-SG (L3 CORE - SIEGE) - AVEC ACL INTERNES
! =====

hostname S1-SG
no ip domain-lookup
ip routing
standby version 2

vlan 10
  name SERVICES
vlan 20
  name PERSONNEL
vlan 30
  name VIDEO
vlan 40
  name PRODUCTION
vlan 50
  name WIFI_STAFF
vlan 60
  name WIFI_GUEST
vlan 99
  name ADMIN
vlan 100
  name TRANSIT

ip domain-name siege.local
ip name-server 10.10.10.10

username admin privilege 15 secret admin
crypto key generate rsa modulus 1024
ip ssh version 2

ip access-list standard MGMT_SSH
  permit 10.10.99.0 0.0.0.15

line vty 0 4
  login local
  transport input ssh
  access-class MGMT_SSH in

service timestamps log datetime msec
logging host 10.10.10.10
logging trap informational
logging on
logging source-interface vlan 99

! =====
! ACL INTERNES (INBOUND SUR SVI)
! =====

ip access-list extended ACL_V10_SERVICES_IN
```



```

remark --- Services : autoriser LAN uniquement (PAS Internet) ---
permit ip 10.10.10.0 0.0.0.255 10.10.0.0 0.0.255.255
permit ip 10.10.10.0 0.0.0.255 10.20.0.0 0.0.255.255
deny ip 10.10.10.0 0.0.0.255 any log

ip access-list extended ACL_V20_PERSONNEL_IN
permit udp any eq 68 any eq 67
permit udp any eq 67 any eq 68
permit udp 10.10.20.0 0.0.0.255 host 10.10.10.10 eq 53
permit tcp 10.10.20.0 0.0.0.255 host 10.10.10.10 eq 53
deny ip 10.10.20.0 0.0.0.255 10.10.99.0 0.0.0.15
deny ip 10.10.20.0 0.0.0.255 10.20.99.0 0.0.0.15
deny ip 10.10.20.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.10.40.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.10.60.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.20.30.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.20.40.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.20.60.0 0.0.0.255
permit ip 10.10.20.0 0.0.0.255 10.10.10.0 0.0.0.255
permit ip 10.10.20.0 0.0.0.255 any

ip access-list extended ACL_V30_VIDEO_IN
permit udp any eq 68 any eq 67
permit udp any eq 67 any eq 68
permit udp 10.10.30.0 0.0.0.255 host 10.10.10.10 eq 53
permit tcp 10.10.30.0 0.0.0.255 host 10.10.10.10 eq 53
deny ip 10.10.30.0 0.0.0.255 10.10.99.0 0.0.0.15
deny ip 10.10.30.0 0.0.0.255 10.20.99.0 0.0.0.15
deny ip 10.10.30.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.30.0 0.0.0.255 10.10.20.0 0.0.0.255
deny ip 10.10.30.0 0.0.0.255 10.10.40.0 0.0.0.255
deny ip 10.10.30.0 0.0.0.255 10.10.50.0 0.0.0.255
deny ip 10.10.30.0 0.0.0.255 10.10.60.0 0.0.0.255
deny ip 10.10.30.0 0.0.0.255 10.20.20.0 0.0.0.255
deny ip 10.10.30.0 0.0.0.255 10.20.40.0 0.0.0.255
deny ip 10.10.30.0 0.0.0.255 10.20.50.0 0.0.0.255
deny ip 10.10.30.0 0.0.0.255 10.20.60.0 0.0.0.255
permit ip 10.10.30.0 0.0.0.255 10.10.10.0 0.0.0.255
permit ip 10.10.30.0 0.0.0.255 any

ip access-list extended ACL_V40_PROD_IN
permit udp any eq 68 any eq 67
permit udp any eq 67 any eq 68
permit udp 10.10.40.0 0.0.0.255 host 10.10.10.10 eq 53
permit tcp 10.10.40.0 0.0.0.255 host 10.10.10.10 eq 53
deny ip 10.10.40.0 0.0.0.255 10.10.99.0 0.0.0.15
deny ip 10.10.40.0 0.0.0.255 10.20.99.0 0.0.0.15
deny ip 10.10.40.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.40.0 0.0.0.255 10.10.20.0 0.0.0.255
deny ip 10.10.40.0 0.0.0.255 10.10.30.0 0.0.0.255
deny ip 10.10.40.0 0.0.0.255 10.10.50.0 0.0.0.255
deny ip 10.10.40.0 0.0.0.255 10.10.60.0 0.0.0.255
deny ip 10.10.40.0 0.0.0.255 10.20.20.0 0.0.0.255

```

```

deny ip 10.10.40.0 0.0.0.255 10.20.30.0 0.0.0.255
deny ip 10.10.40.0 0.0.0.255 10.20.50.0 0.0.0.255
deny ip 10.10.40.0 0.0.0.255 10.20.60.0 0.0.0.255
permit ip 10.10.40.0 0.0.0.255 10.10.10.0 0.0.0.255
permit ip 10.10.40.0 0.0.0.255 any

ip access-list extended ACL_V50_STAFF_IN
permit udp any eq 68 any eq 67
permit udp any eq 67 any eq 68
permit udp 10.10.50.0 0.0.0.255 host 10.10.10.10 eq 53
permit tcp 10.10.50.0 0.0.0.255 host 10.10.10.10 eq 53
deny ip 10.10.50.0 0.0.0.255 10.10.99.0 0.0.0.15
deny ip 10.10.50.0 0.0.0.255 10.20.99.0 0.0.0.15
deny ip 10.10.50.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.50.0 0.0.0.255 10.10.30.0 0.0.0.255
deny ip 10.10.50.0 0.0.0.255 10.10.40.0 0.0.0.255
deny ip 10.10.50.0 0.0.0.255 10.10.60.0 0.0.0.255
deny ip 10.10.50.0 0.0.0.255 10.20.30.0 0.0.0.255
deny ip 10.10.50.0 0.0.0.255 10.20.40.0 0.0.0.255
deny ip 10.10.50.0 0.0.0.255 10.20.60.0 0.0.0.255
permit ip 10.10.50.0 0.0.0.255 10.10.10.0 0.0.0.255
permit ip 10.10.50.0 0.0.0.255 10.10.20.0 0.0.0.255
permit ip 10.10.50.0 0.0.0.255 any

ip access-list extended ACL_V60_GUEST_IN
permit udp any eq 68 any eq 67
permit udp any eq 67 any eq 68
permit udp 10.10.60.0 0.0.0.255 host 10.10.10.10 eq 53
permit tcp 10.10.60.0 0.0.0.255 host 10.10.10.10 eq 53
deny ip 10.10.60.0 0.0.0.255 10.10.0.0 0.0.255.255
deny ip 10.10.60.0 0.0.0.255 10.20.0.0 0.0.255.255
deny ip 10.10.60.0 0.0.0.255 10.10.100.0 0.0.0.7
permit ip 10.10.60.0 0.0.0.255 any

ip access-list extended ACL_V99_ADMIN_IN
remark --- ADMIN : LAN uniquement (PAS Internet) ---
permit ip 10.10.99.0 0.0.0.15 10.10.0.0 0.0.255.255
permit ip 10.10.99.0 0.0.0.15 10.20.0.0 0.0.255.255
deny ip 10.10.99.0 0.0.0.15 any log

ip access-list extended ACL_V100_TRANSIT_IN
deny ip 10.10.20.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.30.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.40.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.50.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.60.0 0.0.0.255 10.10.100.0 0.0.0.7
permit ip any any

! =====
! SVIs + HSRP + ACL IN
! =====
interface vlan 100
ip address 10.10.100.5 255.255.255.248
ip access-group ACL_V100_TRANSIT_IN in

```

```

no shutdown

interface vlan 10
ip address 10.10.10.2 255.255.255.0
standby 10 ip 10.10.10.1
standby 10 priority 110
standby 10 preempt
ip access-group ACL_V10_SERVICES_IN in
no shutdown

interface vlan 20
ip address 10.10.20.2 255.255.255.0
standby 20 ip 10.10.20.1
standby 20 priority 110
standby 20 preempt
ip helper-address 10.10.10.10
ip access-group ACL_V20_PERSONNEL_IN in
no shutdown

interface vlan 30
ip address 10.10.30.2 255.255.255.0
standby 30 ip 10.10.30.1
standby 30 priority 110
standby 30 preempt
ip helper-address 10.10.10.10
ip access-group ACL_V30_VIDEO_IN in
no shutdown

interface vlan 40
ip address 10.10.40.2 255.255.255.0
standby 40 ip 10.10.40.1
standby 40 priority 110
standby 40 preempt
ip helper-address 10.10.10.10
ip access-group ACL_V40_PROD_IN in
no shutdown

interface vlan 50
ip address 10.10.50.2 255.255.255.0
standby 50 ip 10.10.50.1
standby 50 priority 110
standby 50 preempt
ip helper-address 10.10.10.10
ip access-group ACL_V50_STAFF_IN in
no shutdown

interface vlan 60
ip address 10.10.60.2 255.255.255.0
standby 60 ip 10.10.60.1
standby 60 priority 110
standby 60 preempt
ip helper-address 10.10.10.10
ip access-group ACL_V60_GUEST_IN in
no shutdown

```

```

interface vlan 99
 ip address 10.10.99.2 255.255.255.240
 standby 99 ip 10.10.99.1
 standby 99 priority 110
 standby 99 preempt
 ip access-group ACL_V99_ADMIN_IN in
 no shutdown

ip route 0.0.0.0 0.0.0.0 10.10.100.4

interface gi1/0/1
 description Vers R1-SG g0/1 (VLAN100)
 switchport mode access
 switchport access vlan 100
 no shutdown

interface gi1/0/11
 description Vers S3-SG (TRUNK)
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,99,100
 no shutdown

```

Configuration de S2-SG

```

! =====
! S2-SG (L3 CORE - SIEGE) - AVEC ACL INTERNES
! =====
hostname S2-SG
no ip domain-lookup
ip routing
standby version 2

vlan 10
 name SERVICES
vlan 20
 name PERSONNEL
vlan 30
 name VIDEO
vlan 40
 name PRODUCTION
vlan 50
 name WIFI_STAFF
vlan 60
 name WIFI_GUEST
vlan 99
 name ADMIN
vlan 100
 name TRANSIT

ip domain-name siege.local

```

```

ip name-server 10.10.10.10

username admin privilege 15 secret admin
crypto key generate rsa modulus 1024
ip ssh version 2

ip access-list standard MGMT_SSH
permit 10.10.99.0 0.0.0.15

line vty 0 4
login local
transport input ssh
access-class MGMT_SSH in

service timestamps log datetime msec
logging host 10.10.10.10
logging trap informational
logging on
logging source-interface vlan 99

! =====
! ACL INTERNES (identiques S1-SG)
! =====

ip access-list extended ACL_V10_SERVICES_IN
permit ip 10.10.10.0 0.0.0.255 10.10.0.0 0.0.255.255
permit ip 10.10.10.0 0.0.0.255 10.20.0.0 0.0.255.255
deny ip 10.10.10.0 0.0.0.255 any log

ip access-list extended ACL_V20_PERSONNEL_IN
permit udp any eq 68 any eq 67
permit udp any eq 67 any eq 68
permit udp 10.10.20.0 0.0.0.255 host 10.10.10.10 eq 53
permit tcp 10.10.20.0 0.0.0.255 host 10.10.10.10 eq 53
deny ip 10.10.20.0 0.0.0.255 10.10.99.0 0.0.0.15
deny ip 10.10.20.0 0.0.0.255 10.20.99.0 0.0.0.15
deny ip 10.10.20.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.10.40.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.10.60.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.20.30.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.20.40.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.20.60.0 0.0.0.255
permit ip 10.10.20.0 0.0.0.255 10.10.10.0 0.0.0.255
permit ip 10.10.20.0 0.0.0.255 any

ip access-list extended ACL_V30_VIDEO_IN
permit udp any eq 68 any eq 67
permit udp any eq 67 any eq 68
permit udp 10.10.30.0 0.0.0.255 host 10.10.10.10 eq 53
permit tcp 10.10.30.0 0.0.0.255 host 10.10.10.10 eq 53
deny ip 10.10.30.0 0.0.0.255 10.10.99.0 0.0.0.15
deny ip 10.10.30.0 0.0.0.255 10.20.99.0 0.0.0.15
deny ip 10.10.30.0 0.0.0.255 10.10.100.0 0.0.0.7

```

```

deny ip 10.10.30.0 0.0.0.255 10.10.20.0 0.0.0.255
deny ip 10.10.30.0 0.0.0.255 10.10.40.0 0.0.0.255
deny ip 10.10.30.0 0.0.0.255 10.10.50.0 0.0.0.255
deny ip 10.10.30.0 0.0.0.255 10.10.60.0 0.0.0.255
deny ip 10.10.30.0 0.0.0.255 10.20.20.0 0.0.0.255
deny ip 10.10.30.0 0.0.0.255 10.20.40.0 0.0.0.255
deny ip 10.10.30.0 0.0.0.255 10.20.50.0 0.0.0.255
deny ip 10.10.30.0 0.0.0.255 10.20.60.0 0.0.0.255
permit ip 10.10.30.0 0.0.0.255 10.10.10.0 0.0.0.255
permit ip 10.10.30.0 0.0.0.255 any

ip access-list extended ACL_V40_PROD_IN
permit udp any eq 68 any eq 67
permit udp any eq 67 any eq 68
permit udp 10.10.40.0 0.0.0.255 host 10.10.10.10 eq 53
permit tcp 10.10.40.0 0.0.0.255 host 10.10.10.10 eq 53
deny ip 10.10.40.0 0.0.0.255 10.10.99.0 0.0.0.15
deny ip 10.10.40.0 0.0.0.255 10.20.99.0 0.0.0.15
deny ip 10.10.40.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.40.0 0.0.0.255 10.10.20.0 0.0.0.255
deny ip 10.10.40.0 0.0.0.255 10.10.30.0 0.0.0.255
deny ip 10.10.40.0 0.0.0.255 10.10.50.0 0.0.0.255
deny ip 10.10.40.0 0.0.0.255 10.10.60.0 0.0.0.255
deny ip 10.10.40.0 0.0.0.255 10.20.20.0 0.0.0.255
deny ip 10.10.40.0 0.0.0.255 10.20.30.0 0.0.0.255
deny ip 10.10.40.0 0.0.0.255 10.20.50.0 0.0.0.255
deny ip 10.10.40.0 0.0.0.255 10.20.60.0 0.0.0.255
permit ip 10.10.40.0 0.0.0.255 10.10.10.0 0.0.0.255
permit ip 10.10.40.0 0.0.0.255 any

ip access-list extended ACL_V50_STAFF_IN
permit udp any eq 68 any eq 67
permit udp any eq 67 any eq 68
permit udp 10.10.50.0 0.0.0.255 host 10.10.10.10 eq 53
permit tcp 10.10.50.0 0.0.0.255 host 10.10.10.10 eq 53
deny ip 10.10.50.0 0.0.0.255 10.10.99.0 0.0.0.15
deny ip 10.10.50.0 0.0.0.255 10.20.99.0 0.0.0.15
deny ip 10.10.50.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.50.0 0.0.0.255 10.10.30.0 0.0.0.255
deny ip 10.10.50.0 0.0.0.255 10.10.40.0 0.0.0.255
deny ip 10.10.50.0 0.0.0.255 10.10.60.0 0.0.0.255
deny ip 10.10.50.0 0.0.0.255 10.20.30.0 0.0.0.255
deny ip 10.10.50.0 0.0.0.255 10.20.40.0 0.0.0.255
deny ip 10.10.50.0 0.0.0.255 10.20.60.0 0.0.0.255
permit ip 10.10.50.0 0.0.0.255 10.10.10.0 0.0.0.255
permit ip 10.10.50.0 0.0.0.255 10.10.20.0 0.0.0.255
permit ip 10.10.50.0 0.0.0.255 any

ip access-list extended ACL_V60_GUEST_IN
permit udp any eq 68 any eq 67
permit udp any eq 67 any eq 68
permit udp 10.10.60.0 0.0.0.255 host 10.10.10.10 eq 53
permit tcp 10.10.60.0 0.0.0.255 host 10.10.10.10 eq 53
deny ip 10.10.60.0 0.0.0.255 10.10.0.0 0.0.255.255

```

```

deny ip 10.10.60.0 0.0.0.255 10.20.0.0 0.0.255.255
deny ip 10.10.60.0 0.0.0.255 10.10.100.0 0.0.0.7
permit ip 10.10.60.0 0.0.0.255 any

ip access-list extended ACL_V99_ADMIN_IN
permit ip 10.10.99.0 0.0.0.15 10.10.0.0 0.0.255.255
permit ip 10.10.99.0 0.0.0.15 10.20.0.0 0.0.255.255
deny ip 10.10.99.0 0.0.0.15 any log

ip access-list extended ACL_V100_TRANSIT_IN
deny ip 10.10.20.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.30.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.40.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.50.0 0.0.0.255 10.10.100.0 0.0.0.7
deny ip 10.10.60.0 0.0.0.255 10.10.100.0 0.0.0.7
permit ip any any

! =====
! SVIs + HSRP + ACL IN
! =====

interface vlan 100
ip address 10.10.100.6 255.255.255.248
ip access-group ACL_V100_TRANSIT_IN in
no shutdown

interface vlan 10
ip address 10.10.10.3 255.255.255.0
standby 10 ip 10.10.10.1
standby 10 priority 90
standby 10 preempt
ip access-group ACL_V10_SERVICES_IN in
no shutdown

interface vlan 20
ip address 10.10.20.3 255.255.255.0
standby 20 ip 10.10.20.1
standby 20 priority 90
standby 20 preempt
ip helper-address 10.10.10.10
ip access-group ACL_V20_PERSONNEL_IN in
no shutdown

interface vlan 30
ip address 10.10.30.3 255.255.255.0
standby 30 ip 10.10.30.1
standby 30 priority 90
standby 30 preempt
ip helper-address 10.10.10.10
ip access-group ACL_V30_VIDEO_IN in
no shutdown

interface vlan 40
ip address 10.10.40.3 255.255.255.0
standby 40 ip 10.10.40.1

```

```

standby 40 priority 90
standby 40 preempt
ip helper-address 10.10.10.10
ip access-group ACL_V40_PROD_IN in
no shutdown

interface vlan 50
ip address 10.10.50.3 255.255.255.0
standby 50 ip 10.10.50.1
standby 50 priority 90
standby 50 preempt
ip helper-address 10.10.10.10
ip access-group ACL_V50_STAFF_IN in
no shutdown

interface vlan 60
ip address 10.10.60.3 255.255.255.0
standby 60 ip 10.10.60.1
standby 60 priority 90
standby 60 preempt
ip helper-address 10.10.10.10
ip access-group ACL_V60_GUEST_IN in
no shutdown

interface vlan 99
ip address 10.10.99.3 255.255.255.240
standby 99 ip 10.10.99.1
standby 99 priority 90
standby 99 preempt
ip access-group ACL_V99_ADMIN_IN in
no shutdown

ip route 0.0.0.0 0.0.0.0 10.10.100.4

interface gil/0/1
description Vers R2-SG g0/1 (VLAN100)
switchport mode access
switchport access vlan 100
no shutdown

interface gil/0/12
description Vers S3-SG (TRUNK)
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,99,100
no shutdown

```

```

R1-SG#traceroute 10.20.20.1
Type escape sequence to abort.
Tracing the route to 10.20.20.1
 0  1  172.16.20.2      1 msec    0 msec    0 msec

```



```

ASM2-SG#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
                0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

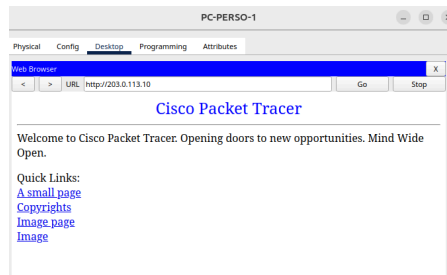
Console logging: level debugging, 46 messages logged, xml disabled,
                filtering disabled
Monitor logging: level debugging, 46 messages logged, xml disabled,
                filtering disabled
Buffer logging: level debugging, 0 messages logged, xml disabled,
                filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped
Trap logging: level informational, 46 message lines logged
Logging to 10.10.10.10 (udp port 514, audit disabled,
                authentication disabled, encryption disabled, link up),
                46 message lines logged,
                0 message lines rate-limited,
                0 message lines dropped-by-MD,
                xml disabled, sequence number disabled
                filtering disabled

```



SAE-303-INFRA (goat).pkt

QOS :

```

R1-SC#show policy-map interface GigabitEthernet0/0
GigabitEthernet0/0

Service-policy output: PM_WAN_OUT

Class-map: CM_VIDEO (match-any)
  0 packets, 0 bytes
  5 minute offered rate 0 bps, drop rate 0 bps
  Match: access-group name QOS_VIDEO_MATCH
    0 packets, 0 bytes
    5 minute rate 0 bps
  Queueing
    Strict Priority
    Output Queue: Conversation 264
    Bandwidth 40 (%)
    Bandwidth 400000 (kbps) Burst 10000000 (Bytes)
    (pkts matched/bytes matched) 0/0
    (total drops/bytes drops) 0/0

Class-map: class-default (match-any)
  3 packets, 805 bytes
  5 minute offered rate 20 bps, drop rate 0 bps
  Match: any
  Queueing
    Flow Based Fair Queueing
    Maximum number of Hashed Queues 256
    Bandwidth 750000 (kbps)Max Threshold 64 (packets)
    (total queued/total drops/no-buffer drops) 0/0/0

R1-SG#show policy-map interface GigabitEthernet0/1
GigabitEthernet0/1

Service-policy output: PM_WAN_OUT

Class-map: CM_VIDEO (match-any)
  0 packets, 0 bytes
  5 minute offered rate 0 bps, drop rate 0 bps
  Match: access-group name QOS_VIDEO_MATCH
    0 packets, 0 bytes
    5 minute rate 0 bps
  Queueing
    Strict Priority
    Output Queue: Conversation 264
    Bandwidth 40 (%)
    Bandwidth 400000 (kbps) Burst 10000000 (Bytes)
    (pkts matched/bytes matched) 0/0
    (total drops/bytes drops) 0/0

Class-map: class-default (match-any)
  21 packets, 901 bytes
  5 minute offered rate 20 bps, drop rate 0 bps
  Match: any
  Queueing
    Flow Based Fair Queueing
    Maximum number of Hashed Queues 256
    Bandwidth 750000 (kbps)Max Threshold 64 (packets)
    (total queued/total drops/no-buffer drops) 0/0/0

```

Voila les resultats :

```

-----
R1-SG#show policy-map interface GigabitEthernet0/1
GigabitEthernet0/1

Service-policy output: PM_WAN_OUT

Class-map: CM_VIDEO (match-any)
 0 packets, 0 bytes
 5 minute offered rate 0 bps, drop rate 0 bps
 Match: access-group name QOS_VIDEO_MATCH
 0 packets, 0 bytes
 5 minute rate 0 bps
 Queueing
  Strict Priority
  Output Queue: Conversation 264
  Bandwidth 40 (%)
  Bandwidth 400000 (kbps) Burst 10000000 (Bytes)
  (pkts matched/bytes matched) 0/0
  (total drops/bytes drops) 0/0

Class-map: class-default (match-any)
 4549 packets, 296468 bytes
 5 minute offered rate 2946 bps, drop rate 0 bps
 Match: any
 Queueing
  Flow Based Fair Queueing
  Maximum number of Hashed Queues 256
  Bandwidth 750000 (kbps)Max Threshold 64 (packets)
  (total queued/total drops/no-buffer drops) 0/0/0

```

SAE-303-INFRA (goat.v2).pkt

(masterclass)